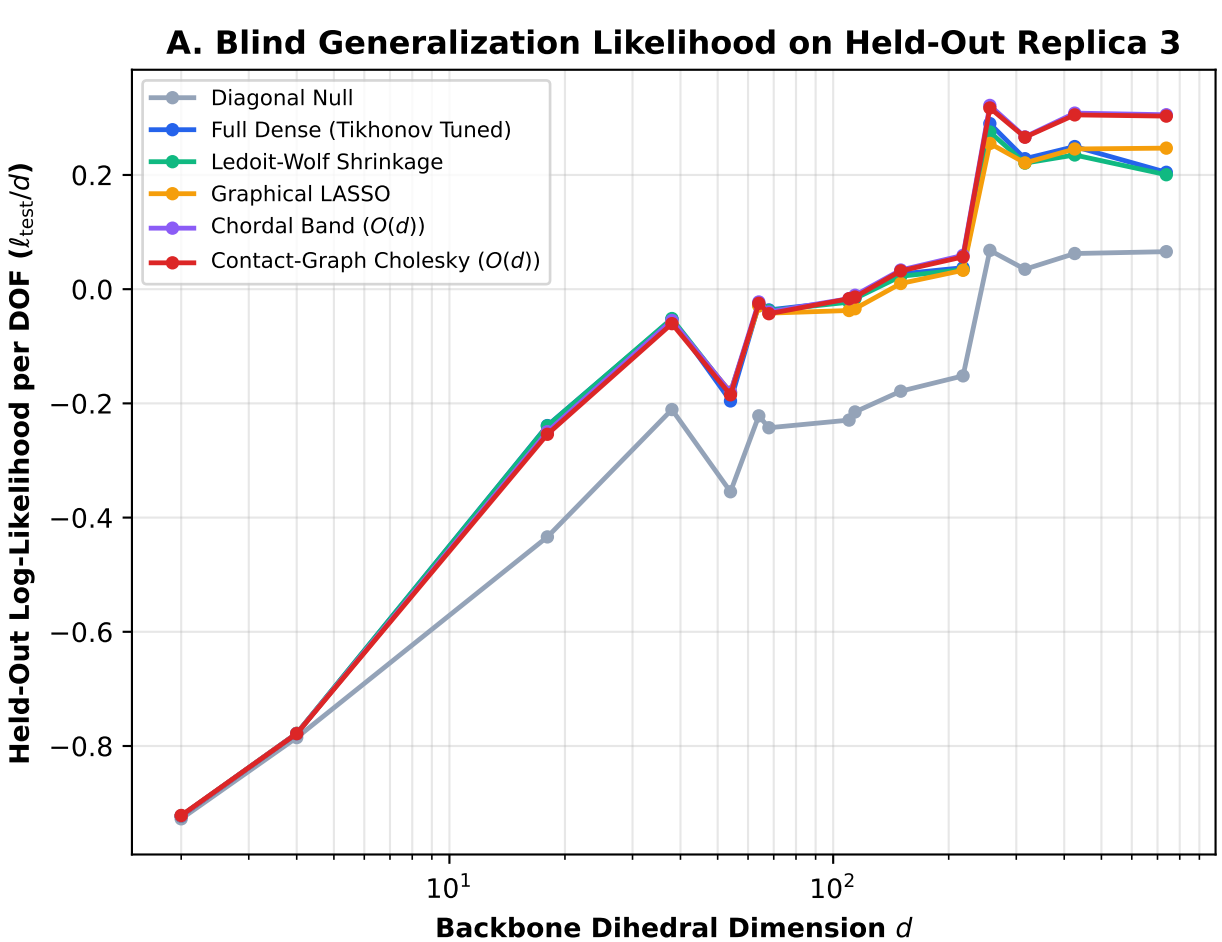
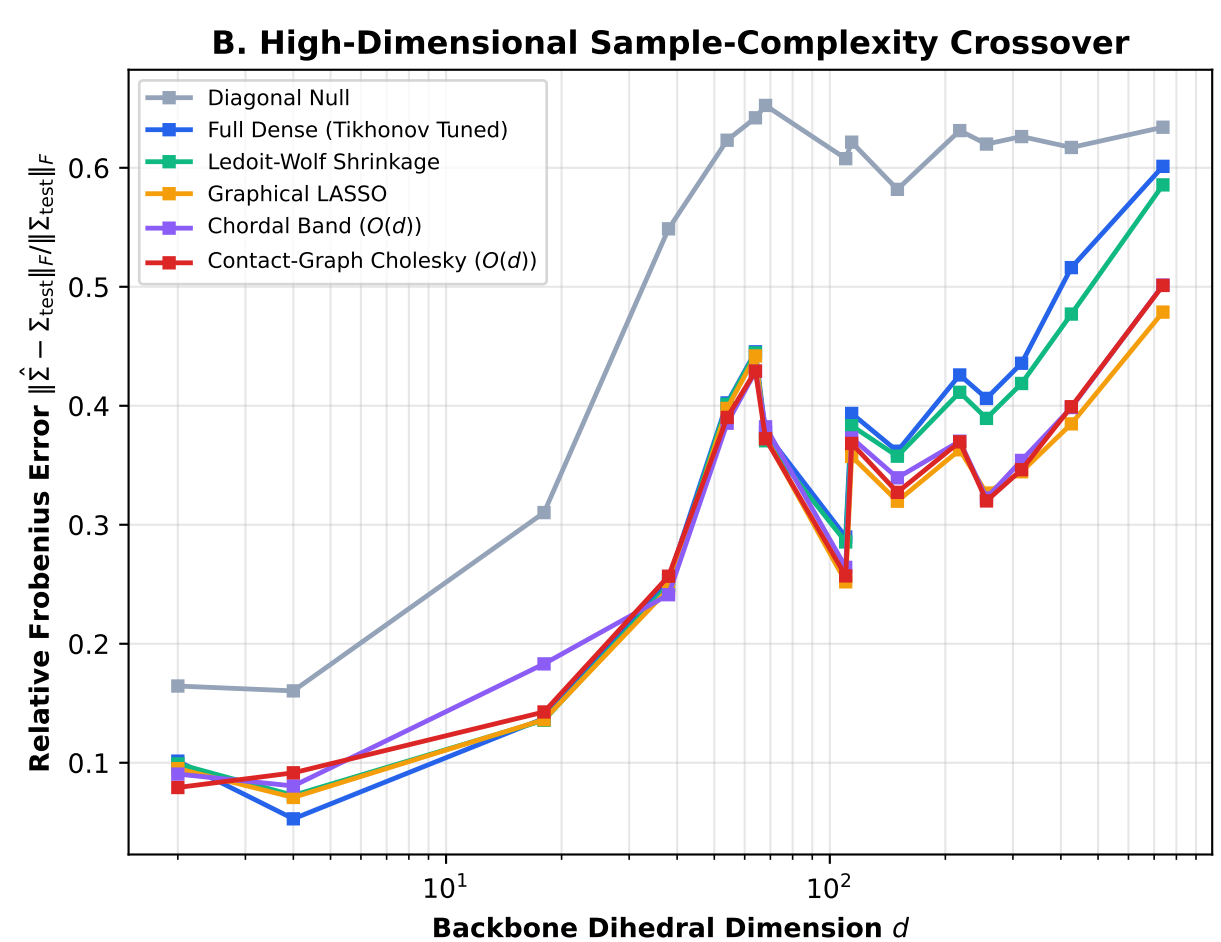
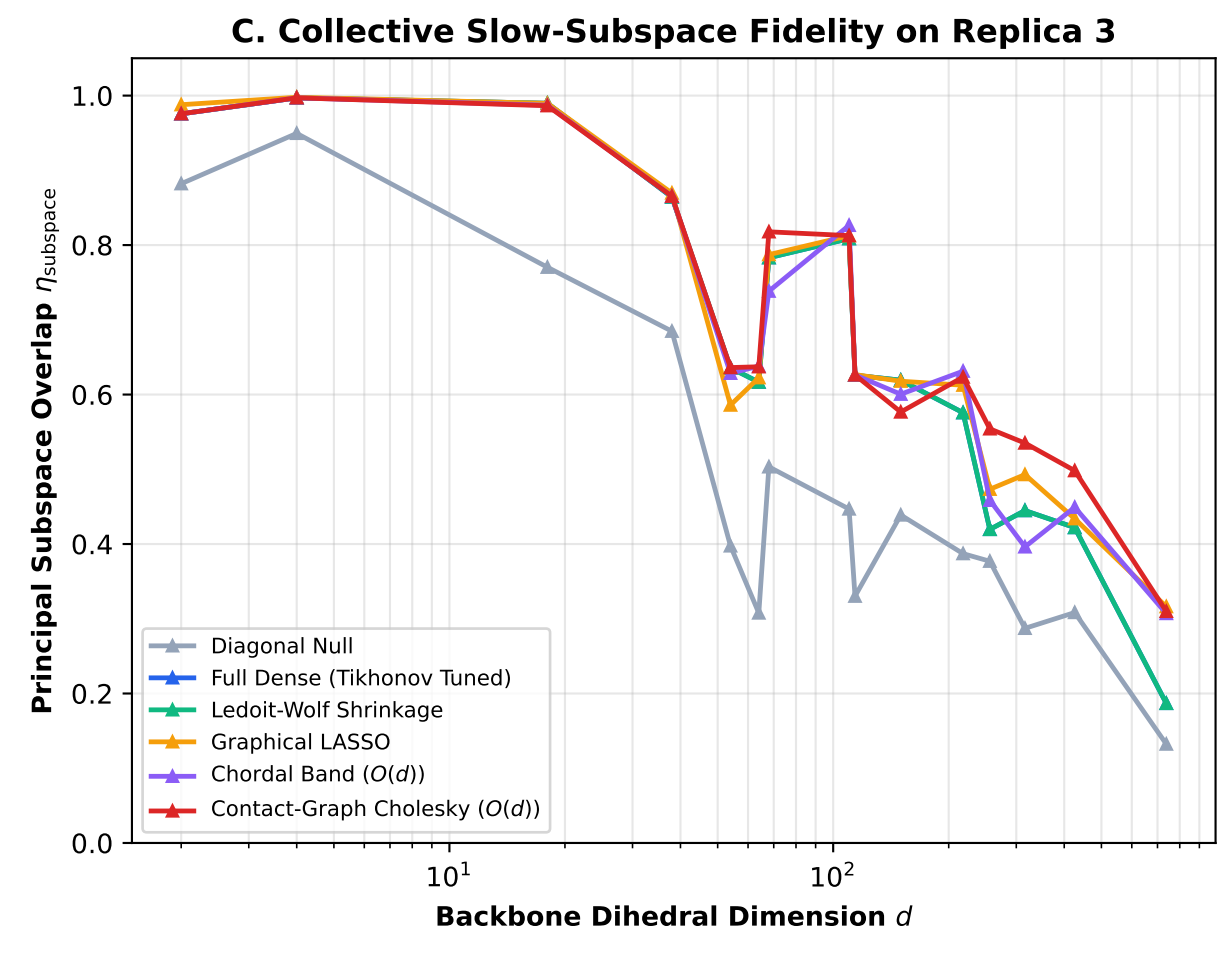
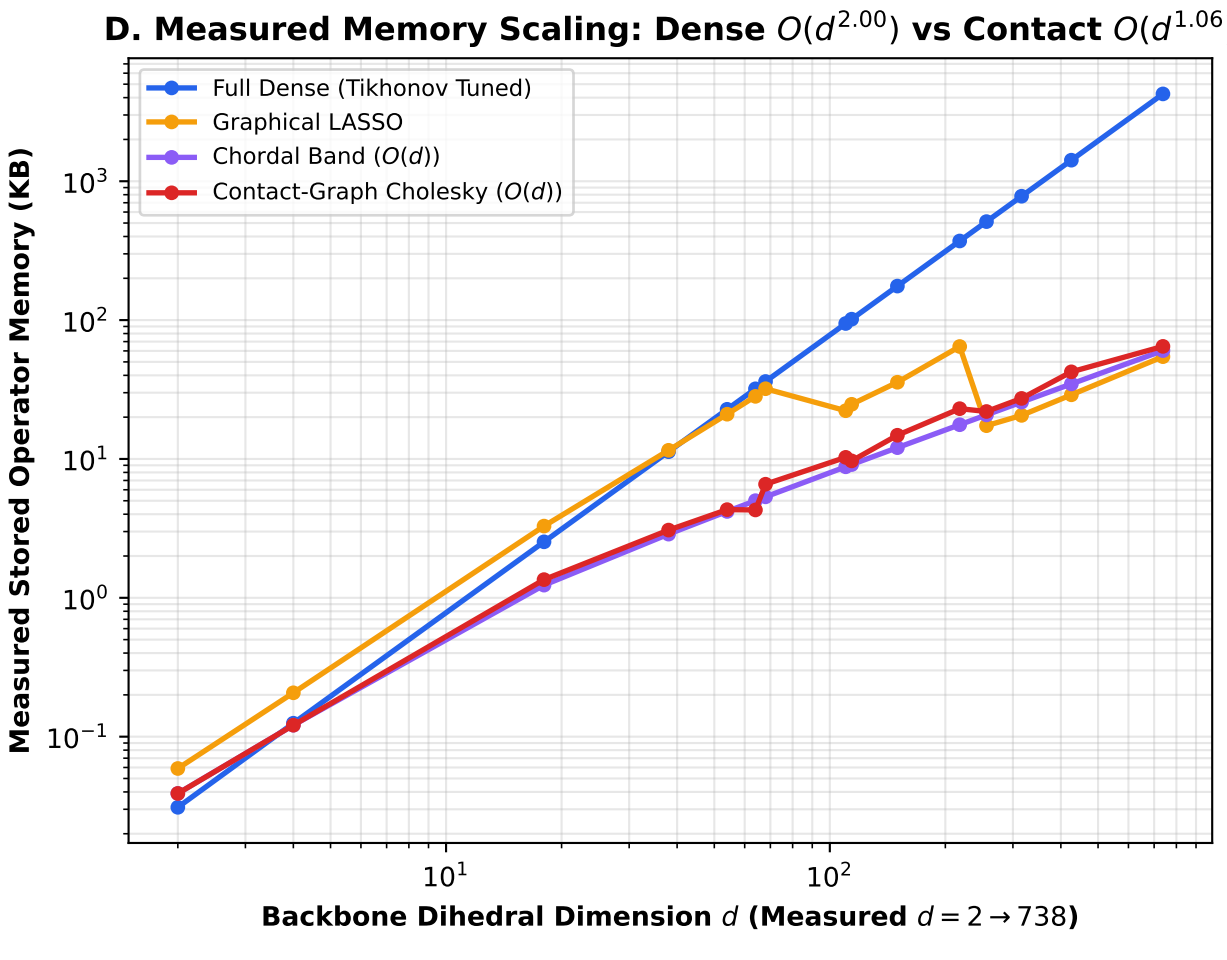

A. Blind Generalization Likelihood on Held-Out Replica 3
Held-Out Log-Likelihood per DOF (ℓ_{test}/d)
Backbone Dihedral Dimension d
Diagonal Null
Full Dense (Tikhonov Tuned)
Ledoit-Wolf Shrinkage
Graphical LASSO
Chordal Band ($O(d)$)
Contact-Graph Cholesky ($O(d)$)


B. High-Dimensional Sample-Complexity Crossover
Relative Frobenius Error $\|\hat{\Sigma} - \Sigma_{\text{test}}\|_F / \|\Sigma_{\text{test}}\|_F$
Backbone Dihedral Dimension d
Diagonal Null
Full Dense (Tikhonov Tuned)
Ledoit-Wolf Shrinkage
Graphical LASSO
Chordal Band ($O(d)$)
Contact-Graph Cholesky ($O(d)$)


C. Collective Slow-Subspace Fidelity on Replica 3
Principal Subspace Overlap η_{subspace}
Backbone Dihedral Dimension d
Diagonal Null
Full Dense (Tikhonov Tuned)
Ledoit-Wolf Shrinkage
Graphical LASSO
Chordal Band ($O(d)$)
Contact-Graph Cholesky ($O(d)$)


D. Measured Memory Scaling: Dense $O(d^{2.00})$ vs Contact $O(d^{1.06})$
Measured Stored Operator Memory (KB)
Backbone Dihedral Dimension d (Measured $d = 2 \rightarrow 738$)
Full Dense (Tikhonov Tuned)
Graphical LASSO
Chordal Band ($O(d)$)
Contact-Graph Cholesky ($O(d)$)